# Assignment 3: Flights of New York

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## Exercise 1

i. How many rows and columns does this dataset have?

There are 19 columns and 336,776 rows.

ii. What does a single row in this dataset represent?

A single flight.

iii. What is the difference between the information contained in the arr_time and sched_arr_time columns? (Take a look at the column descriptions)

arr_time is the actual time that the plane arrived, sched_arr_time is when it was suppoed to arrive.

iv. Airplanes are reused across many different flights. Which column(s) would be helpful to use in identifying individual airplanes?

Yes, they were reused. The tailnum column helps with this.

## Exercise 2

```
flights %>%
  select(year, month)
```

```
## # A tibble: 336,776 x 2
##     year month
##    <int> <int>
##  1  2013     1
##  2  2013     1
##  3  2013     1
##  4  2013     1
##  5  2013     1
##  6  2013     1
##  7  2013     1
##  8  2013     1
##  9  2013     1
## 10  2013     1
## # i 336,766 more rows
```

### Exercise 3

```
flights %>%
  select(year: day)
```

```
## # A tibble: 336,776 x 3
##     year month   day
##    <int> <int> <int>
##  1  2013     1     1
##  2  2013     1     1
##  3  2013     1     1
##  4  2013     1     1
##  5  2013     1     1
##  6  2013     1     1
##  7  2013     1     1
##  8  2013     1     1
##  9  2013     1     1
## 10  2013     1     1
## # i 336,766 more rows
```

The colon selects only the columns from year to day.

### Exercise 4

```
flights %>%
  arrange(air_time, distance) %>%
  select(air_time, distance)
```

```
## # A tibble: 336,776 x 2
##    air_time distance
##       <dbl>    <dbl>
##  1       20      116
##  2       20      116
##  3       21       80
##  4       21       80
##  5       21       94
##  6       21      116
##  7       21      116
##  8       21      116
##  9       21      116
## 10       21      116
## # i 336,766 more rows
```

i. Does it look like both the air_time and distance columns were sorted?

No, they weren't.

ii. Which column was sorted first?

air_time

iii. What happens if you reverse the order you specify the columns in arrange()?

Then distance gets arranged first.

### Exercise 5

```
flights %>%
  arrange(desc(dep_delay)) %>%
  select(dep_delay, carrier, flight)
```

```
## # A tibble: 336,776 x 3
##    dep_delay carrier flight
##        <dbl> <chr>    <int>
##  1      1301 HA          51
##  2      1137 MQ        3535
##  3      1126 MQ        3695
##  4      1014 AA         177
##  5      1005 MQ        3075
##  6       960 DL        2391
##  7       911 DL        2119
##  8       899 DL        2007
##  9       898 DL        2047
## 10       896 AA         172
## # i 336,766 more rows
```

The longest departure delay was carrier HA and flight number 51

### Exercise 6

```
flights %>%
  mutate(average_speed = distance / (air_time / 60)) %>%
  select(year: dep_delay, average_speed)
```

```
## # A tibble: 336,776 x 7
##     year month   day dep_time sched_dep_time dep_delay average_speed
##    <int> <int> <int>    <int>          <int>     <dbl>         <dbl>
##  1  2013     1     1      517            515         2          370.
##  2  2013     1     1      533            529         4          374.
##  3  2013     1     1      542            540         2          408.
##  4  2013     1     1      544            545        -1          517.
##  5  2013     1     1      554            600        -6          394.
##  6  2013     1     1      554            558        -4          288.
##  7  2013     1     1      555            600        -5          404.
##  8  2013     1     1      557            600        -3          259.
##  9  2013     1     1      557            600        -3          405.
## 10  2013     1     1      558            600        -2          319.
## # i 336,766 more rows
```

i. Where does the new column you just computed show up in the dataset and what is the name

of this new column?

It is the last column, and its called average_speed.

ii. What part of the code is controlling the name of the new column?

Where it equals to the columns that its going to mutate.

### Exercise 7

```
flights %>%
  mutate(dep_time_hour = dep_time %/% 100,
         dep_time_minute = dep_time %% 100,
         dep_time_minutes_midnight = (dep_time_hour * 60) + dep_time_minute) %>%
  select(dep_time_hour: dep_time_minutes_midnight)
```

```
## # A tibble: 336,776 x 3
##    dep_time_hour dep_time_minute dep_time_minutes_midnight
##            <dbl>           <dbl>                     <dbl>
##  1             5              17                       317
##  2             5              33                       333
##  3             5              42                       342
##  4             5              44                       344
##  5             5              54                       354
##  6             5              54                       354
##  7             5              55                       355
##  8             5              57                       357
##  9             5              57                       357
## 10             5              58                       358
## # i 336,766 more rows
```

### Exercise 8

```
flights %>%
  filter(arr_delay < 0, carrier == "AA") %>%
  select(year: arr_time ,arr_delay: carrier)
```

```
## # A tibble: 20,769 x 9
##     year month   day dep_time sched_dep_time dep_delay arr_time arr_delay
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>     <dbl>
##  1  2013     1     1      606            610        -4      858       -12
##  2  2013     1     1      628            630        -2     1137        -3
##  3  2013     1     1      656            659        -3      949       -10
##  4  2013     1     1      659            700        -1     1008        -7
##  5  2013     1     1      712            715        -3     1023       -12
##  6  2013     1     1      739            745        -6      918       -12
##  7  2013     1     1      753            755        -2     1056       -14
##  8  2013     1     1      803            810        -7      903       -22
##  9  2013     1     1      840            845        -5     1311       -39
```

```
## 10  2013     1     1      940            945        -5     1119       -11
## # i 20,759 more rows
## # i 1 more variable: carrier <chr>
```

### Exercise 9

```
flights %>%
  group_by(carrier) %>%
  summarize(average_arr_delay = mean(arr_delay, na.rm = TRUE)) %>%
  arrange(desc(average_arr_delay))
```

```
## # A tibble: 16 x 2
##    carrier average_arr_delay
##    <chr>               <dbl>
##  1 F9                 21.9
##  2 FL                 20.1
##  3 EV                 15.8
##  4 YV                 15.6
##  5 OO                 11.9
##  6 MQ                 10.8
##  7 WN                  9.65
##  8 B6                  9.46
##  9 9E                  7.38
## 10 UA                  3.56
## 11 US                  2.13
## 12 VX                  1.76
## 13 DL                  1.64
## 14 AA                  0.364
## 15 HA                 -6.92
## 16 AS                 -9.93
```

i. Which airline carrier had the longest arrival delays on average?

F9 has the longest delay, 21.92.

ii. Which airline carrier had the shortest arrival delays on average?

AS has the shortest delay, -9.93.

```
flights %>%
  group_by(carrier) %>%
  summarize(
    average_arr_delay = mean(arr_delay, na.rm = TRUE),
    average_dep_delay = mean(dep_delay, na.rm = TRUE)
  )
```

```
## # A tibble: 16 x 3
##    carrier average_arr_delay average_dep_delay
##    <chr>               <dbl>             <dbl>
##  1 9E                  7.38              16.7
##  2 AA                  0.364              8.59
```

```
##  3 AS                -9.93              5.80
##  4 B6                 9.46             13.0
##  5 DL                 1.64              9.26
##  6 EV                15.8              20.0
##  7 F9                21.9              20.2
##  8 FL                20.1              18.7
##  9 HA                -6.92              4.90
## 10 MQ                10.8              10.6
## 11 OO                11.9              12.6
## 12 UA                 3.56             12.1
## 13 US                 2.13              3.78
## 14 VX                 1.76             12.9
## 15 WN                 9.65             17.7
## 16 YV                15.6              19.0
```

### Exercise 10

```
late_flights_to_miami <- flights %>%
  filter(dest == "MIA" & arr_delay > 0) %>%
  select(arr_delay, carrier)
```

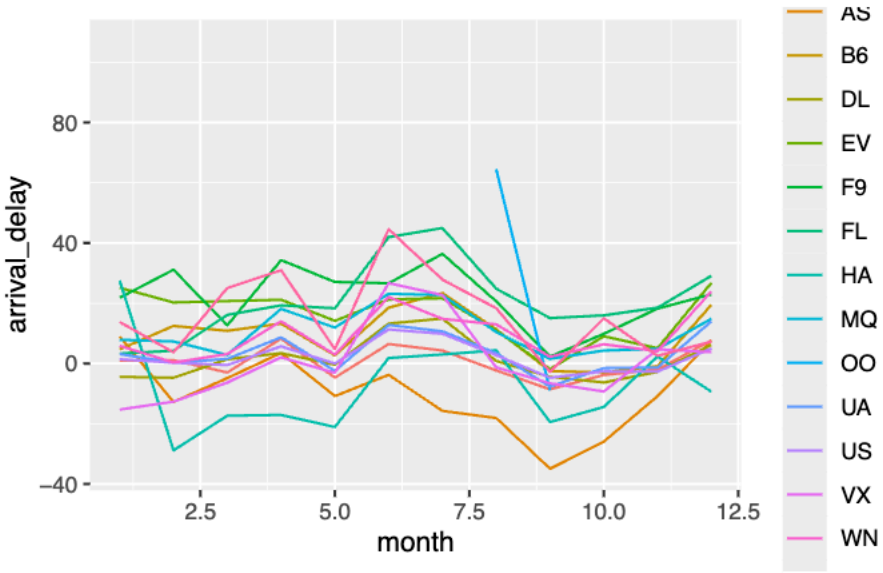
### Exercise 11

#### New Data Set

```
monthly_delays <- flights %>%
  group_by(month, carrier) %>%
  summarize(
    arrival_delay = mean(arr_delay, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  spread(carrier, arrival_delay) %>%
  select(-`9E`)
```

```
pivoted_monthly_delays <- monthly_delays %>%
  pivot_longer(cols = -month, names_to = "airline", values_to = "arrival_delay")
```

```
library(ggplot2)
qplot(x = month,
      y = arrival_delay,
      color = airline,
      geom = "line",
      data = pivoted_monthly_delays)
```


arrival_delay
80
40
0
−40
2.5
5.0
7.5
10.0
12.5
month
AS
B6
DL
EV
F9
FL
HA
MQ
OO
UA
US
VX
WN